# Read the axes before you read the line

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Signed area / ES: área con signo / PT: área com sinal

Height × width = displacement. Area below zero is negative displacement.

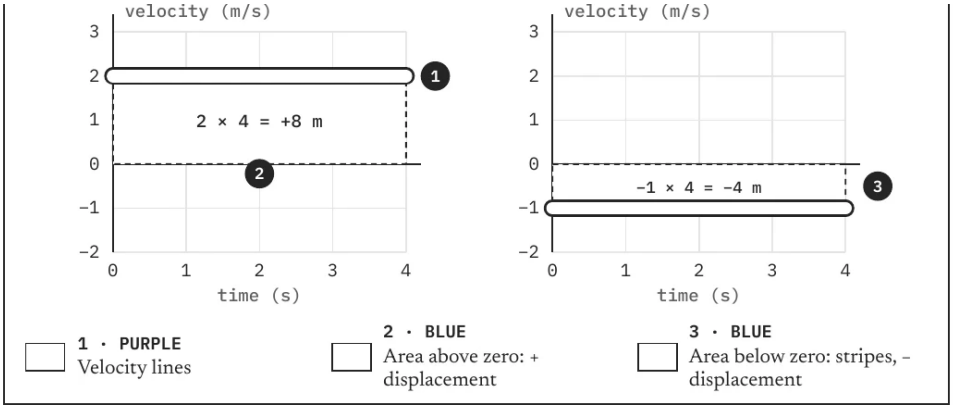
Also on your note page: Velocity–time graph, Constant velocity, Components.

## B READ THE RELATIONSHIP

### Signed area on a velocity–time graph gives displacement.

Displacement = signed area under a velocity–time graph

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### Velocity is +3 m/s for 2 s. Find the displacement and name its direction.

**Given** $v = +3$ m/s $t = 2$ s

**Rule** $\Delta x = v \times t$

**Work** $\Delta x = (3 \text{ m/s})(2 \text{ s})$

**Answer** +6 m 6 m to the right

## E SAVE FOR THE FINAL MEETING

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

# Read the physics: Signed area

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Read the vertical axis first. A flat line on a velocity–time graph means steady velocity. A flat line above zero means the object is moving.

## 02 HOW IT WORKS

Shade the region between velocity and the time axis. For steady velocity, multiply velocity by time. Add areas above zero; subtract the sizes of areas below zero. For distance, add all the sizes.

## 03 WHAT TO CHECK

A diagonal velocity has two components. For an angle measured from +x, use cosine for the x part and sine for the y part.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

Velocity is +3 m/s for 2 s. Find the displacement and name its direction. +6 m 6 m to the right.

## 05 CONNECT TO XJ

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

SAY IT BEFORE YOU WRITE

Which shaded part subtracts from displacement? The area is ___ because it is ___ zero.

# Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

## A FRAYER MODEL

### Signed area / ES: área con signo / PT: área com sinal

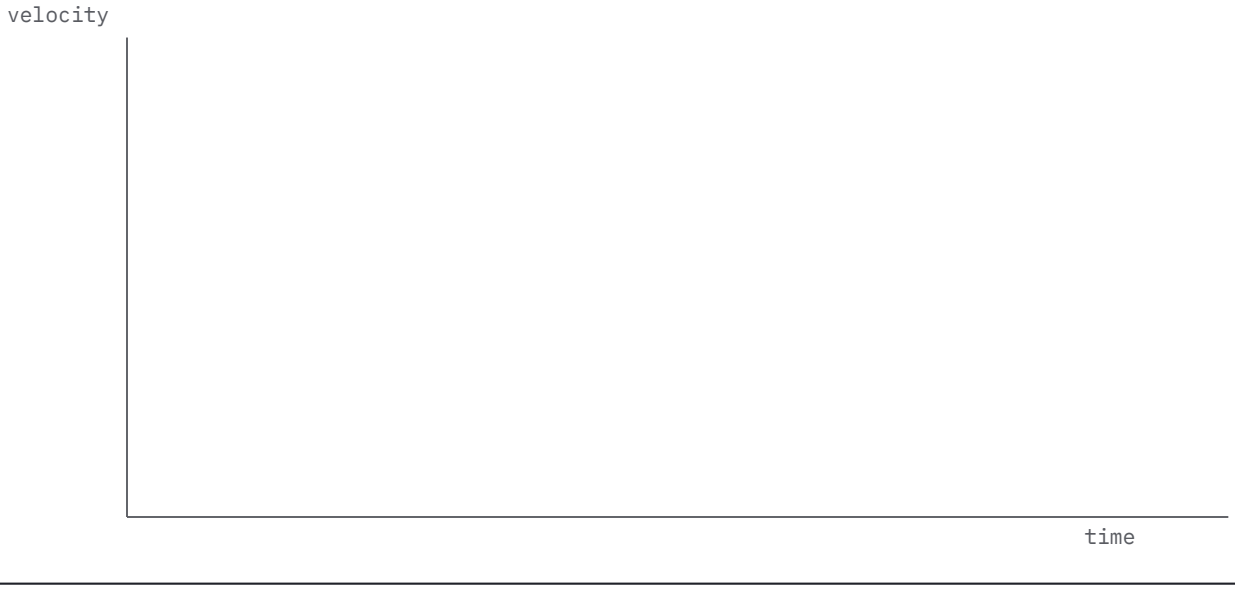
Say it in any language. Draw or use a few words. Use the notes to check your idea.

| Meaning | How can you tell? |
|---|---|
| To me, signed area means … | On a velocity–time graph, how can you tell which area to subtract? I subtract the area that lies ___. |
| **Example or sketch** | **Non-example** |
| I see it when … | This looks close, but … |

## B DRAW WITH YOUR TEACHER

### Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.

Labels you can use: above · below · area.




**Tell a partner:** The area is ___ because it is ___ zero.

# On assignment: Signed area

SECTION E / BACK / FINAL MEETING ONLY Name _____

**A moving camera follows a model at +3 km/s for 2 s, then −1 km/s for 2 s. The editor needs the end location and the total footage distance.**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Tracking engineer**

Shade both parts of a velocity graph. Give the camera operator the final change from start, including direction.

☐ **Replay producer**

Storyboard the outbound and return shots. Tell the editor total distance and net movement; mark which part plays in reverse direction.

**Word bank: above · below · area.**

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

**“We first said ___. Now we ___ because ___.”**